

Saw It or Triggered It: Exploring the Threshold of Implicit and Explicit Interaction for Eye-tracking Technique in Virtual Reality

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ABSTRACT

With eye-tracking techniques, the virtual reality (VR) system can acquire what the user is looking at in the virtual environment (VE). Most previous works focused on eye-tracking interaction in VR with explicit interactions. In this paper, we tried to determine whether the eye-tracking techniques can be applied as implicit interactions. With implicit eye-tracking interactions, users could trigger the story events unconsciously without missing them. The approaches could be interactive techniques for enhancing VR storytelling. We conducted a user study to investigate the threshold of implicit and explicit interaction for the eye-tracking technique in VR. We designed three interfaces: objects lighting, objects movement, and objects sound, for users to judge whether they trigger objects in the VE or not when seeing them. The result provides a parameter in the design guideline for immersive storytelling with the eye-tracking device and brings a new VR storytelling technique.

Keywords: Implicit interaction, explicit interaction, eye-tracking technique, immersive storytelling, virtual reality.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Eye-tracking technologies have been adopted or built-in in the virtual reality (VR) head-mounted display. VR developers can acquire and utilize users' eye gaze as mechanisms to trigger events with the eye-tracking technique. Users would not miss the virtual environment (VE) events, which becomes an essential technique for VR's immersive experience. Prior studies used eye-tracking techniques as explicit interactions in VR. Tanriverdi and Jacob [6] first utilized eye-gaze to select objects in the VR and found it more beneficial than conventional hand-based selection. Sidorakis et al. [5] compared the effect of eye-tracking controllers and the traditional mouse/keyboard input and showed that users make fewer type errors while operating the eye-controlled interface than the standard keyboard during immersive viewing. Piumsombon et al. [3] explored eye gaze techniques for selection and menu applications based on natural eye movements. Probst et al. [4] utilize the user's eye-tracking information to trigger events in the immersive experience. EyeSeeThrough [1] systems unified tool selection and application with eye-tracking techniques in VR, which gave users rapid interaction with context menu interfaces in VR. Through this interaction technique, VR developers can make sure that users would not miss the change of events. Users can also engage in the immersive experience since they interact with the events in the VE.

Nevertheless, in some scenarios, the VR director wants to make users feel that they "just catch up" on VE's events instead of triggering the events. For example, catching the moment of seeing the sunset in the virtual world is greatly different from triggering the sunset. Previous works have utilized diegetic, or non-diegetic

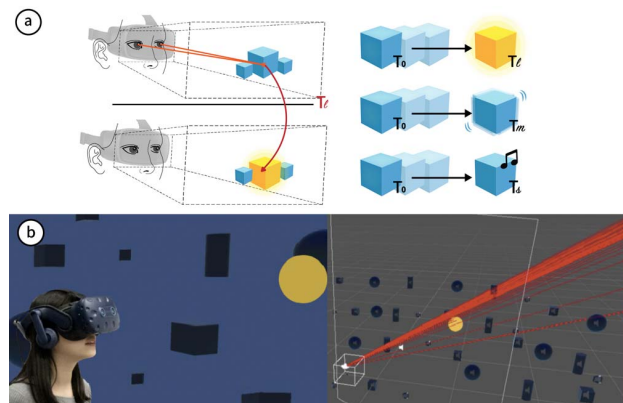


Figure 1: (a) shows the experimental design and three interfaces in our experiment. (b) shows the user in the VR is trying to figure out if she triggers the objects to light by her eye gaze.

cues [2] to implicitly direct users' attention, such as following the firefly by their gazes. These techniques required additional information to drive users' sight. Therefore, the threshold of the implicit and explicit interaction for the eye-tracking technique in VR becomes an interesting issue. By exploring the threshold, we can make sure that users would not miss the events since the events would be triggered by the eye gaze and make users feel the events are just happening in their sight.

This paper investigates how the time difference between the eye trigger and the change of objects affects users' feelings in the VE (as Figure 1) and explores this threshold of implicit and explicit interactions for VR director to have an additional parameter to enhance the users' immersive experiences in VR storytelling.

2 USER STUDY

To determine if different feedback types would affect different result thresholds. We designed three interfaces: objects lighting(OL), objects movement (OM), and objects sound (OS), for users to judge whether they trigger objects in the VE or not when seeing them. 18 participants (7 females, age 22-32, $\bar{x} = 25.4$, $SD = 3.6$) volunteered to take part in the experiment, with interpupillary distance between 62.5 mm and 68.1 mm ($\bar{x} = 64.3$ mm, $SD = 1.61$). All had experience in VR applications.

To investigate the maximum delay time that the users can feel it is themselves who trigger the objects to change status, we adopted the Absolute Detection Threshold (ADT). We utilized the two-up, one-down transformed staircase method and the Yes/No task to estimate an absolute threshold. We started the experiment with an immediate response, 0 sec delay time, which users can feel that it is themselves who trigger the objects to change status. When the users think that the objects were triggered by themselves twice in a row, we add 0.88 sec to the delay time and add the same if they still feel that they can trigger the objects. On the other hand, if the users answer that they cannot feel that the objects were triggered by themselves, we minus 0.88 sec to the delay time. The experiments would end if users answer "No Feel" for 0 sec delay time three times to avoid

negative delay time. We change to add or minus 0.44 sec to the delay time after two reversals. Repeatedly, after four reversals, we add or minus 0.22 sec to the delay time. And add or minus 0.11 sec after six reversals. The experiments end after eight total reversals, and we take the average of the last four as the threshold estimate.

We use HTC VIVE Pro Eye to detect the users' eye gaze in the VE. To start the experiment, the users need to adjust the interpupillary distance on their headset, ensuring that users' eye gaze can be detected precisely. After users click the start button in each interface, they would see a virtual world full of 100 different objects. And only 20 objects can be triggered by the users. The rest 80 objects would be randomly chosen to change the status that corresponds to interface setting status. After 30 sec, the question would pop up, and users could either answer the question immediately or keep observing the objects for the rest of the 30 sec. After 60 sec, users must answer the question without objects in the scene. After users answer the question, the objects would be reset again with new positions, new 20 objects would be selected as triggerable, and new delay time.

3 RESULTS

We computed the absolute detection threshold for each participant from the last four delay time reversals in three interfaces. In the OL interface, participants' thresholds varied between 0.082 and 2.805 sec. In the OM interface, participants' thresholds varied between 0.083 and 2.777 sec. In the OS interface, participants' thresholds varied between 0.083 and 0.8 sec. And we got the mean and standard deviation for three interfaces (As shown in Figure 2). We end up using the mean value minus one standard deviation as the lowest delay time that users think they trigger the objects by themselves when seeing them. The result thresholds are 0.12 sec (OL), 0.13 sec (OM), and 0.03 sec (OS). We conduct the three interfaces with the Two-Factor Without Replication Anova and the Tukey test and found the OS interface has a distinctive difference from the OL and OM interfaces.

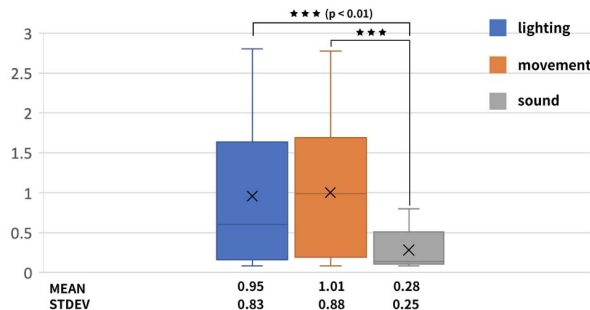


Figure 2: Result thresholds of 18 participants in three interfaces

4 DISCUSSION AND LIMITATION

We found in Figure 2 that the delay time in the first two interfaces, OL and OM, behaved quite the same but are significantly different from the OS interfaces. The possible reason is that both the OL and OM interfaces use only eyes as the input and output of status change during the experiment, which means the trigger mechanism and reaction mechanism are the same, say seeing objects. However, in the OS interface, users need to use both eyes and ears as the input and output of status change during the experiment. That is when the eye gaze triggered objects, the sound was heard by the ears. Therefore, we might say that in the VE, the threshold of implicit and explicit interaction would decrease when the trigger mechanism and the reaction mechanism are different.

In our experiment, we conduct the experiment from explicit to implicit interaction. We figure out the threshold by starting the experiment from 0 delay time, which the users can definitely feel that they trigger the objects when seeing them. Then we concluded

that the users could feel they trigger the objects below the threshold, and above the threshold, the users would feel that the change of objects might not trigger by themselves. However, the threshold value might be different if we start the experiment with a long delay time and decrease the delay time when users answer they cannot feel they trigger the objects. Users absolutely cannot feel that objects were triggered by themselves when seeing them. Therefore, we should experiment with the same participants in the future but from implicit interaction to explicit interaction and compare the two thresholds figuring out a more precise threshold value.

5 CONCLUSION

The result thresholds could be a parameter of the story writing guideline when arranging events in VR storytelling. For example, when various virtual world events need to be triggered by users, eye-tracking is a critical device to get users' attention. The story director could give every event order with the importance of different response times, such as the most important event with the immediate response time and the least important event with the delay time above the threshold. We can make sure that users thoroughly experience the whole VR storytelling because none of the events would show up outside users' sight. Besides, with the feelings of implicit and explicit interactions, users would not be distracted by the less important events when they show up after a delayed response time because they would not feel those events are triggered by themselves. On the other hand, users can immerse themselves in more important events because they know they trigger the events. Hence, our threshold value would be a powerful mechanism that tracks the users' eye direction to trigger the events. With a specific delay time, the users would feel that the events would happen naturally without missing them, making users more immersive in this natural VR storytelling. In conclusion, this threshold could be a useful mechanism to enhance immersive VR storytelling.

ACKNOWLEDGMENT

This research was supported by the Ministry of Science and Technology of Taiwan under the grant MOST 109-2218-E-002-038-

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